Ashraf Hany

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Education

2021 - 2025 BSc. in Computer Science, **MSA University**, Egypt **(Dual Degree Program)** **CGPA: 3.81/4**
First Class Honors

2021 - 2025 BSc. in Computer Science, **Greenwich University**, United Kingdom
First Class Honors

Achievements

Third Place Engineering Track - 20th NUGRF ***Spring 2025***

- Awarded Engineering Track's third place in the most competitive national undergraduate competition (NU UGRF) held at Egypt's Nile University for building and developing an ESP32 handheld game console (Gameboy) from scratch.

Best Undergraduate Project ***Spring 2023***

- Awarded (Sophomore Best Undergraduate Project) for publishing a research paper in an IEEE conference during the 2nd year of my undergraduate studies.

MSA CPC Founding Community Leader ***[Link]*** Fall 2023

- Founded MSA Competitive Programming Community and participated at the ECPC 2023 & ECPC 2024 with 21 & 28 teams, respectively.

Second Place at NU Quantum-AI Hackathon ***[Link]*** ***Summer 2024***

- The challenge centered on a Raman shift spectroscopy dataset from a Nature paper. We were tasked with preprocessing the data to reduce 1389 features for encoding into a VQC (Variational Quantum Classifier), in addition to building and optimizing the VQC model itself using Qiskit.

Experience

Teaching Assistant | **MSA University** August 2025 – Present

- Taught five groups of 30 students each the **Design & Analysis of Algorithms** course, covering sorting, recursion, dynamic programming, and complexity analysis.
- Taught the **Embedded Systems** course, including microcontroller programming, interfacing, and real-time applications.
- Designed and delivered lab sessions, assignments, and problem-solving tutorials to reinforce theoretical concepts.
- Developed and presented **educational animations and visualizations** using **Manim** to enhance comprehension of complex algorithms and system operations.
- Assisted in evaluating students through lab performance, project presentations, quizzes, and final examinations.
- Provided individual guidance and academic support to students struggling with programming logic and hardware implementation.

Research Intern | **Nile University** August - September 2024

- Researched quantum machine learning (QML) models in noisy environments to improve fault tolerance.
- Researched and analyzed techniques for optimizing QML performance under realistic noise conditions.

Freelancer | **Upwork** August 2024 – Present

- Designed and developed educational and scientific animations using **Manim** for clear visual communication.
- Conducted **AI research**, including implementation and evaluation of machine learning and quantum machine learning models.
- Collected, cleaned, and preprocessed data from diverse sources to build high-quality datasets.
- Documented workflows covering data collection, preprocessing, modeling, and visualization.

Research Intern | **Nile University** August - September 2024

- Researched quantum machine learning (QML) models in noisy environments to improve fault tolerance.
- Researched and analyzed techniques for optimizing QML performance under realistic noise conditions.

Quantum Machine Learning Researcher | **MSA University** February 2024 – Present

- Utilizing quantum machine learning algorithms using Pennylane for a research project aimed at developing a QML classifier model and testing it on multiple datasets.
- Learning Quantum Computing.

Junior Teaching Assistant | **MSA University** Fall 2023

- Volunteered as a Junior Teaching Assistant for the courses:

`MTH103`	Discrete Math
`CS213`	Multimedia Programming
`CS232`	Algorithms and Data Structures
`CS102`	Fundamentals Of Computing II

Intern | **National Bank of Egypt** July 2023 – August 2023

- Developed a python script to check for variants of a domain name using symbol-oriented permutation.
- Back-seated the top industry professionals on Web Pentration Tests, Network Pentration Tests, Threat intelligence, Patching, VA, Incidents Handling, Phishing.

PUBLICATIONS

Hany, Ashraf; Ramadan, Enjy; Akl, Amr; Atia, Ayman. *The Effect of Using Tangible User Interfaces Compared to Traditional Learning for Teaching Programming in Higher Education: An Experimental Study*. In: *2023 Intelligent Methods, Systems, and Applications (IMSA)*, 2023, pp. 514-519. DOI: `10.1109/IMSA58542.2023.10217780`

Ramadan, Enjy; et al. *Integrating Interactive 3D Applications for Enhanced Learning in Crown Preparations: A Comparative Experimental Study*. In: *2025 International Conference on Activity and Behavior Computing (ABC)*, Al Ain, United Arab Emirates, 2025, pp. 1-10. DOI: `10.1109/ABC64332.2025.11118293`

Hany, Ashraf; Mohammed, A. *Deep Learning for Microscopic Image Segmentation in Multiple Myeloma: A Comparative Study*. In: *2025 Intelligent Methods, Systems, and Applications (IMSA)*, Giza, Egypt, 2025, pp. 256-262. DOI: `10.1109/IMSA65733.2025.11167795`

Hany, Ashraf; Gomma, W. H. *Comprehensive Benchmark: Comparing Classical, Recurrent, and Transformer-Based Models for Sentiment Analysis Across Text Length Strata*. In: *2025 Intelligent Methods, Systems, and Applications (IMSA)*, Giza, Egypt, 2025, pp. 487-494. DOI: `10.1109/IMSA65733.2025.11166950`

PROJECTS

Quantum Random Walk for Procedural Generation*[Link]* | *Python, PennyLane, Procedural Generation* Fall 2024

- Developed a quantum computing approach to procedural terrain generation using quantum walk algorithms on a 2D grid, demonstrating practical applications of quantum principles in game development and computational graphics.
- Implemented quantum circuits using PennyLane with 2 qubits to represent directional movement (X and Y axes), applying Hadamard gates to create superposition states for randomized terrain generation.
- Designed and executed a quantum walk simulation over 200 steps on a 10×10 grid, mapping quantum measurement outcomes to directional movements (up, down, left, right) with proper boundary wrapping.
- Research is still being conducted at MSA University exploring quantum procedural generation techniques for gaming applications, showcasing the potential of quantum computing in creative computational tasks beyond traditional optimization problems.

ESP32-S3 Gameboy*[Link]* | *C, Retro-Go, Embedded Systems* Spring 2025

- Developed a fully custom handheld gaming console based on the ESP32-S3 microcontroller.
- Integrated a 3.2-inch display, custom-designed button PCB, MAX9837 audio DAC, and a 3D-printed ergonomic case.
- Optimized and deployed Retro-Go firmware to emulate multiple classic gaming systems including NES, Game Boy, and Sega platforms at full speed on the ESP32-S3
- Demonstrated a cost-effective DIY solution showcasing how open-source hardware and software can deliver premium retro gaming experiences. Project Shortlisted at NUGRF 20th Edition.

Quantum Varitional Classifier | *Qiskit, Python* Summer 24 - Hackathon

- Secured second place in Egypt's first Quantum-AI Hackathon organized by Nile University.
- Worked on a Raman shift spectroscopy dataset from a recent Nature paper.
- Preprocessed the data to reduce 1389 features for encoding into a Variational Quantum Classifier (VQC).
- Developed and optimized the VQC model using Qiskit's RealAmplitudes ansatz, ZZFeatureMap, COBYLA optimizer, and Sampler.

Real-Time Object Tracking & Following using Raspberry Pi 5 and YOLO | *Computer Vision* Spring 25 - Course Project

- Developed a real-time object detection and tracking system using YOLO (You Only Look Once) algorithm on Raspberry Pi 5 hardware

- Implemented computer vision techniques for continuous object following with camera control and servo motor integration
- Optimized model performance for edge computing, achieving real-time processing speeds suitable for embedded systems
- Integrated OpenCV and PyTorch libraries for image processing and neural network inference on ARM architecture

Utilizing SVM and Decision Trees in Breast Cancer Detection | *Scikit-learn, Matplotlib, Numpy* Spring 24 - Course Project

- Developed and evaluated breast cancer detection models using Python and scikit-learn, achieving a consistent 96% accuracy with Decision Tree and Support Vector Machine algorithms.
- Applied advanced machine learning techniques including ensemble methods and feature engineering to refine model performance, emphasizing robust generalization on unseen data.
- Executed model validation with cross-validation and various performance metrics to ensure reliability and guide future enhancements with hyperparameter tuning and algorithm comparison

Tangible User Interface | *Java, TUIO* Spring 22 - Research Project

- Developed a tangible user interface using the TUIO library that can be interacted with physically using 3D Printed objects.
- Directed a research project in Human-Computer Interaction (HCI). As the first author and project lead.
- Explored the application of Tangible User Interfaces (TUIs) in higher education programming, with the goal of assessing their effectiveness in comparison to traditional learning methods.
- Discussed this research with the Former Minister of Education in Egypt, Dr. Tarek Shawki.

Multimedia Game Development | *.NET, C#, NAudio* Spring 23 - Course Project

- Designed and developed **Escape the Castle,** A 2D action game set in a medieval castle environment. Utilized `.NET` and `C#` for game development.
- Implemented immersive audio experiences using `NAudio`. Created diverse sound effects for characters and weapons.

SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), NoSQL (MongoDB), LaTeX , JavaScript, HTML/CSS
Frameworks: Django, Larvel,Bootstrap ,jQuery
Developer Tools: Git, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse
Libraries: Pygame, Networkx, PyTorch, NumPy, Matplotlib, PennyLane
Mathematics: Calculus, Discrete Math, Linear Algebra, Numerical Analysis/Methods, Coding Theory
Algorithmics: Various Design Strategies (Greedy, Divide & Conquer, Dynamic Programming etc.)

EXTRACURRICULAR COURSES

Harvard CS50 *[CERTIFICATE]* (Harvard, edX)
Harvard CS50 AI With Python *[CERTIFICATE]* (Harvard, edx)
Machine Learning Specialization (Andrew Ng) *[CERTIFICATE]* (Coursera, Deeplearning.Ai, Stanford)
Microsoft Azure AI *[CERTIFICATE]* (Microsoft)
Microsoft Azure Fundamentals *[CERTIFICATE]* (Microsoft)